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Article

Integrated Opto-Synaptic IGZO Transistors for Image Recognition Fabricated at Room Temperature

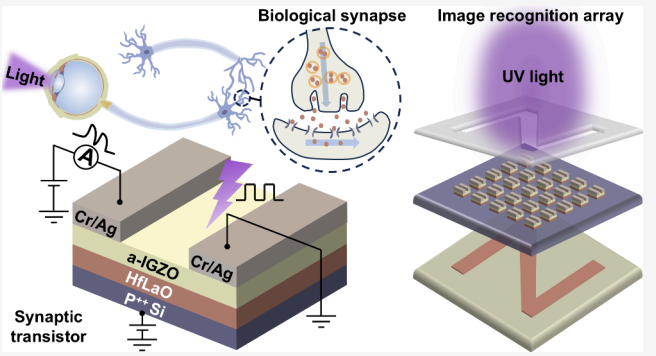
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ABSTRACT: At room temperature, high-κ HfLaO is adopted as the gate dielectric to fabricate amorphous InGaZnO (a-IGZO) optical synaptic thin-film transistors (TFTs), for which plasma treatments are conducted on the HfLaO dielectric in O_2 and a-IGZO in Ar, respectively, namely, OPT/APT-TFTs. Consequently, high-performance a-IGZO TFTs are obtained with a high carrier mobility of 20.8 $cm^2/V{\cdot}s$, a high I_{on}/I_{off} ratio of 3.2×10^6, and a small subthreshold swing (SS) of 0.25 V/dec. As compared to the pristine TFTs, the photocurrent of the OPT/APT-TFTs under a 365 nm ultraviolet (UV) light is significantly raised three times up to 1.4 μA. Meanwhile, the current decay percentage after irradiation removal is reduced from 98% down to 36% within 60 s, indicating an enhanced persistent-photoconductivity (PPC) effect. Accordingly, various optical synaptic plasticities are obtained based on which a simulated neuronal network with a high 93.22% accuracy is achieved to recognize MNIST handwritten digits. Moreover, both neurotransmitter and neuromodulator behaviors are concurrently emulated in a single device through exploiting the native three-terminal structure of the TFT. Importantly, an artificial visual nervous system is successfully constructed by integrating the a-IGZO optoelectronic TFTs for image recognition.



KEYWORDS: *a-IGZO TFTs, optical synaptic plasticities, persistent-photoconductivity effect, high-κ dielectric, image recognition*

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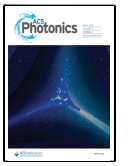
1. INTRODUCTION

With the continuous development of information technology, such as artificial intelligence, massive amounts of data are generated from complex tasks to create an era of big data. As a result, the collection, processing, storage, and transmission of numerous data demand higher performances of intelligent computing systems. However, a bottleneck widely exists in traditional von Neumann computing systems because the memory unit and processor unit are physically separated, limiting the data transmission speed and power efficiency. Inspired by biological neural networks like the human brain consisting of synapses, innovative neuromorphic computing systems have attracted considerable attention to meet the rapidly growing demand for computing power and efficiency.[1−3]

In a biological neural system, external information can be perceived and converted by neurons to generate chemical signals, which can be transmitted among synapses through neurotransmitters, e.g., glutamate for the excitatory postsynaptic current (EPSC) and GABA (γ-aminobutyric acid) for the inhibitory postsynaptic current (IPSC).[4−6] Importantly, the activation of neurotransmitters has been reported to be significantly influenced by surrounding neuromodulators like astrocytes in the biological neuron system.[7,8] Astrocytes can release various gliotransmitters, such as ATP (adenosine triphosphate) and D-serine, to activate neuronal receptors, enhancing synaptic plasticities.[9] Although traditional silicon-based complementary metal-oxide-semiconductor (CMOS) technology has been utilized to implement synaptic functions, challenges of large-area integration and low power dissipation still remain because 10 silicon transistors are usually required to mimic one synapse.[10] Hence, a single novel device to mimic one synapse has been drawing attention, namely, a synaptic device.[11−14]

As one application of neuromorphic electronics, optical synaptic devices can emulate synaptic behaviors under an optical stimulus, which enables photoneuromorphic computing and artificial visual perception system. As compared to synaptic devices triggered by electrical signals, optical synapses are

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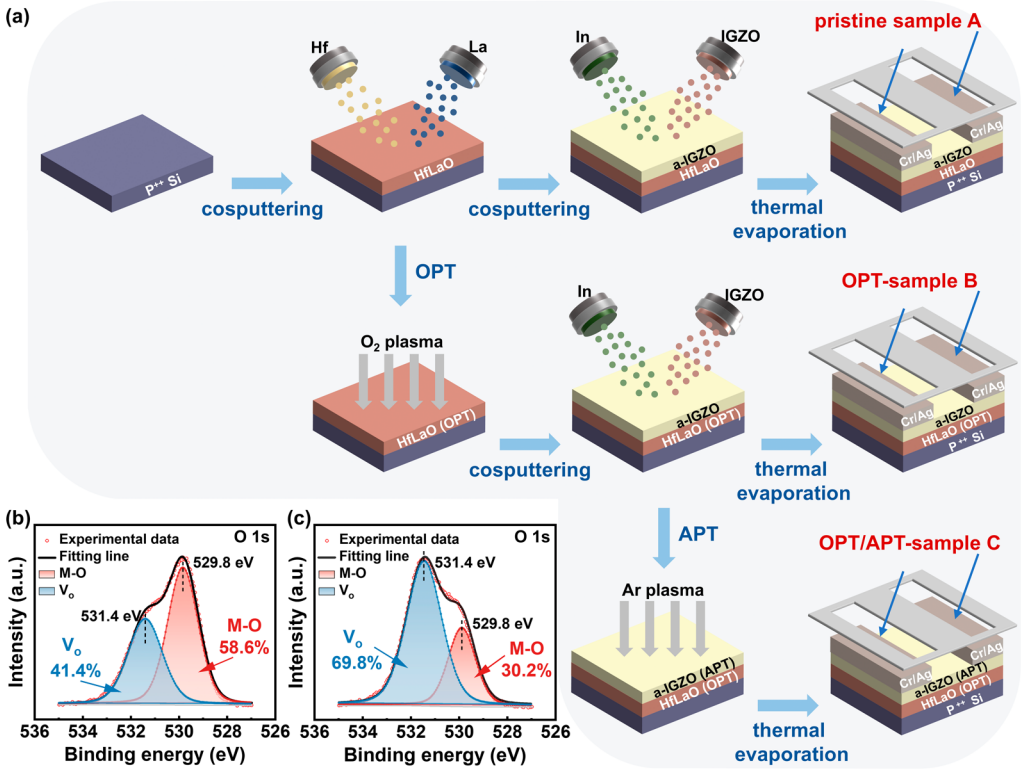


Figure 1. (a) Room-temperature fabrication steps of the a-IGZO optical synaptic TFTs with high-κ HfLaO as the gate dielectric. O 1s XPS spectra of (b) pristine a-IGZO and (c) Ar-plasma-treated (APT) a-IGZO.

expected to be beneficial for high-speed and high-density applications due to the inherent advantages of optical signals, including fast propagation speed, large bandwidth, low crosstalk, and low power consumption.[15−17] Moreover, a high-performance artificial vision perception system can be implemented by optoelectronic synapses, concurrently achieving sensing and memorizing abilities.

In past years, optoelectronic synaptic devices with emerging materials such as two-dimensional materials,[18,19] perovskites,[20,21] helical Nanofiber,[22] and oxide semiconductors[23] have been reported, which can successfully emulate the biological synaptic behavior under various light illumination. Among these, amorphous indium−gallium−zinc oxide (a-IGZO), a famous oxide semiconductor that is widely used in TFTs for panel displays, has attracted much attention due to its high electron mobility, high transparency, low-temperature processing, large-area deposition, and low cost.[24] Besides, a-IGZO shows a selective response to ultraviolet (UV) wavelengths due to a bandgap over 3.0 eV. Importantly, persistent photoconductivity (PPC) behavior can be handily obtained based on its oxygen vacancy-related carrier transport under UV stimulation, which is crucial for enabling optical synaptic behaviors.[25]

Recently, a-IGZO TFTs have been extensively exploited for optical synaptic devices because the three-terminal transistor-type optoelectronic synapses are advantageous for linear conductance switching and long-term operation durability. Notably, the gate-terminal modulation can function as a neuromodulator in biological systems, enhancing the emulation capability of optoelectronic devices. However, the operating voltage of a-IGZO TFTs require further reductions to approach the low power consumption of biological synapses. One feasible way is to adopt high-κ materials as gate dielectrics, which have presented huge successes in Si-based advanced technology nodes.[26] High-κ dielectrics offer excellent gate control, low operating voltages, and mitigation of quantum-tunneling, which can effectively address the limitations of traditional SiO_2.[27,28] As high-κ oxides with a single metal element usually struggle with various oxide defects and hygroscopicity, mixtures of high-κ oxides have been investigated to make them complement each other.[28−30] Moreover, doping, heterojunction structure, process optimization, and postdeposition treatments, like annealing and plasma, are also well-known to improve the performance of TFTs.[31−37]

In this work, a-IGZO optical synaptic TFTs with high-κ HfLaO as the gate dielectric were fabricated by magnetron sputtering at room temperature. The postdeposition plasma treatments with Ar (APT) on a-IGZO and O_2 (OPT) on HfLaO were adopted to improve the interfacial quality for high electrical performances and the PPC effect. The underlying effects of plasma treatments on a-IGZO TFTs were investigated by using X-ray photoelectron spectroscopy (XPS) and atomic force microscopy (AFM). Interestingly, various optical synaptic functions were successfully simulated in the a-IGZO TFTs under light pulses of 365 nm UV, presenting paired-pulse facilitation (PPF), long-term potentiation/depression (LTP/LTD), and transitions from short-term memory (STM) to long-term memory (LTM). Based on the above synaptic parameters, the potentiation and depression characteristics of the a-IGZO synaptic device are adopted as the weight update rule to implement an artificial neural network (ANN), the simulated recognition accuracy of which can reach 93.22% for handwritten digits in the Modified National Institute of Standards and Technology (MNIST) data set. Finally, an optical perception array is successfully

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constructed to recognize and memorize a UV pattern that could be modulated by various gate voltages.

2. RESULTS AND DISCUSSION

Figure 1a shows the room-temperature fabrication processes of the a-IGZO TFTs with high-κ HfLaO as the gate dielectric. According to the fabrication conditions, the samples can be divided into three groups: (a) pristine a-IGZO TFT without the argon and oxygen plasma treatments (APT and OPT), namely, pristine sample A; (b) a-IGZO TFT only with the OPT on HfLaO, namely, OPT-sample B; and (c) a-IGZO TFT with both the OPT on HfLaO and APT on a-IGZO, namely, OPT/APT-sample C. The detailed device fabrication process can be found in the Experimental Section.

XPS and AFM are performed to investigate the properties of the a-IGZO film. As displayed in Figure 1b,c, the XPS O 1s spectra of the a-IGZO can be decomposed into two subpeaks for metal−oxygen bonds (M-O) and oxygen vacancies (V_o), which are located at 529.8 and 531.4 eV, respectively.[38] The binding energies have been calibrated according to the reference of C 1s at 284.8 eV. Owing to the APT, the content ratio of the V_o in a-IGZO is significantly raised from 41.4% to 69.8%, while the M-O is reduced from 58.6% to 30.2%. Besides, a shift of about 1.7 eV to a higher binding energy is observed for the O 1s peak, further suggesting more formations of oxygen vacancies. Meanwhile, the XPS In 3d spectra of the a-IGZO films are shown in Figure S1 (Supporting Information). The spectra can be decomposed into three components located at 443.9, 444.5, and 445.2 eV, which reflect the indium atoms with unsaturated bonding (In_I), indium atoms with saturated bonding (In_{II}), and the indium atoms with neighboring positively charged vacancies (In_{III}) respectively.[39] The In_{III} content is significantly raised from 19.8% to 46.6% by the APT along with significantly reduced In_{II} and In_I content, which suggests an obvious increase in oxygen vacancies. These results imply that more free electrons can be produced due to the donor-type nature of oxygen vacancies in the a-IGZO, favoring the formation of ohmic contacts with source/drain electrodes.[40−42] As depicted by the AFM images in Figure S2 (Supporting Information), the surface of the a-IGZO is effectively smoothened by the APT, as reflected by the reduced root-mean-squared (RMS) values from 0.84 to 0.54 nm. This improved surface morphology is conducive to S/D electrode contacts for high-performance transistors with high mobility.

Additionally, the properties of the high-κ HfLaO dielectric film treated by oxygen plasma are also investigated by XPS and AFM. The oxygen vacancies (V_o) in the HfLaO are effectively suppressed by O_2 plasma treatment along with the significantly increased Hf−O bonds, as reflected by the O 1s XPS spectra in Figure S3 (Supporting Information), which indicates the reduced interfacial defects at the dielectric/channel interface. The content of the La−O bonds decreases, which is believed to contribute to an increase in the number of hydroxyl bonds. This is attributed to the severe hygroscopicity of the La oxide content in the HfLaO film, which can be enhanced due to the surface activation by oxygen plasma treatment.[43,44] As shown in Figure S4, the surface morphologies of the HfLaO films are depicted by AFM (Figure S4, Supporting Information). A smoother surface is obtained for the OPT HfLaO film with a declined RMS value from 0.76 to 0.51 nm due to the sputtering effect of the plasma treatment, which can remove microscopic bulges on the surface. As a result, the quality of the dielectric/channel interface can be effectively improved by the O_2 plasma treatment on the HfLaO dielectric with a smoother surface and resultant fewer oxide defects.

The output characteristics of all the samples are shown in Figure 2a−c, where the on-state current is obviously increased

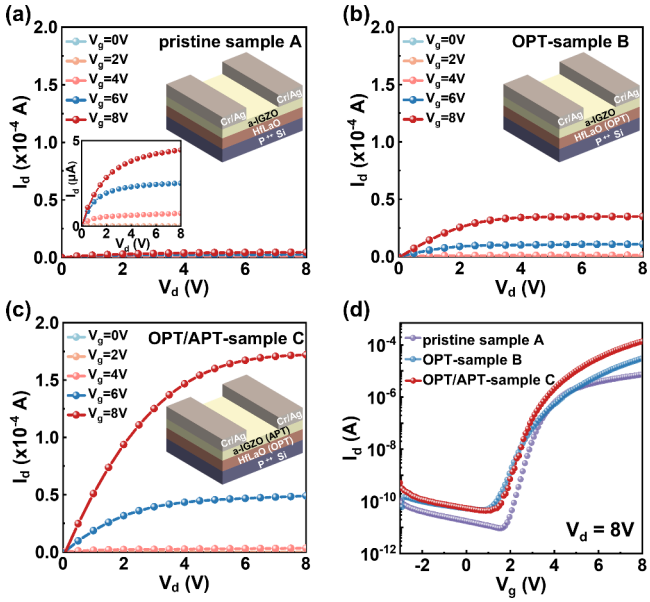


Figure 2. Output characteristics of the a-IGZO TFTs: (a) pristine sample A, (b) OPT-sample B, and (c) OPT/APT-sample C; (d) Transfer characteristics of the pristine sample A, OPT-sample B, and OPT/APT-sample C.

with OPT on HfLaO and APT on a-IGZO. The contact resistance (R_c) of the source and drain electrodes was extracted by a typical transmission line method (TLM), as shown in Figure S5 (Supporting Information).[45] A comparable R_c around 1.2 kΩ is obtained for the pristine sample A and the OPT-sample B due to their same pristine a-IGZO films that were simultaneously deposited. Moreover, the R_c is significantly reduced to 0.2 kΩ for the OPT/APT-sample C because its a-IGZO film was treated by Ar plasma, which induces more donor-like oxygen vacancies to improve the ohmic contact between electrodes and the a-IGZO film.

As shown in Figure 2d, the transfer characteristics of all the samples were measured by sweeping the V_g from −3 to 8 V with a fixed V_d at 8 V, based on which the key electrical parameters are extracted and listed in Table 1. The

Table 1. Main Electrical Parameters of the Fabricated a-IGZO TFTs

sample	μ_{sat} ($cm^2/V{\cdot}s$)	V_{th} (V)	SS (V/dec)	N_t ($cm^{-2}{\cdot}eV^{-1}$)	I_{on}/I_{off}	R_c (kΩ)
A	1.6	2.6	0.32	3.99×10^{12}	8.2×10^5	1.2
B	7.3	2.0	0.26	2.85×10^{12}	6.7×10^5	1.2
C	20.8	2.1	0.25	2.71×10^{12}	3.2×10^6	0.2

corresponding transfer hysteresis characteristics are also shown in Figure S6 (Supporting Information). The pristine sample A exhibits the lowest carrier mobility of 1.6 $cm^2/V{\cdot}s$, which can be dramatically promoted to 20.8 $cm^2/V{\cdot}s$ for the OPT/APT-sample C with both oxygen and Ar plasma treatments. This improvement can be ascribed to the

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smoothened HfLaO film by oxygen plasma treatment, weakening surface roughness scattering.[46] Meanwhile, oxygen vacancy defects at the dielectric/channel interface are also passivated to suppress interfacial scattering, like Coulomb scattering, which can be reflected by the reduced SS and N_t in Table 1. Additionally, the mixture HfLaO high-κ adopted here can effectively passivate interfacial defects that are severely associated with those consisting of single metal element, as reflected in Figure S7 (Supporting Information). Besides, more donor-like oxygen vacancies are induced by extra Ar plasma treatment on a-IGZO to promote the film conductance, which can simultaneously improve the contact between S/D electrode and the a-IGZO to obtain high mobility. This can also be reflected by the correlations of morphological improvement and indium chemical changes with the device performance metrics as shown in Figures S9 and S10 (Supporting Information). Moreover, the threshold voltage was extracted by reading the gate voltage at a drain current of I_d = 10 nA × L/W, where L and W are the channel length and width, respectively.[40,47] Leakage currents can be observed for all devices at negative gate voltage range, which are mainly ascribed to the common gate induced drain leakage effect, namely, GIDL in FETs.[48] With a fixed 8 V V_d, the drain/gate potential difference increases with V_g decreasing from 0 to −3 V, which can cause a promotion of the GIDL.

Figure 3a and b show the typical photoresponse characteristics of the a-IGZO TFTs under 365 nm UV light illumination

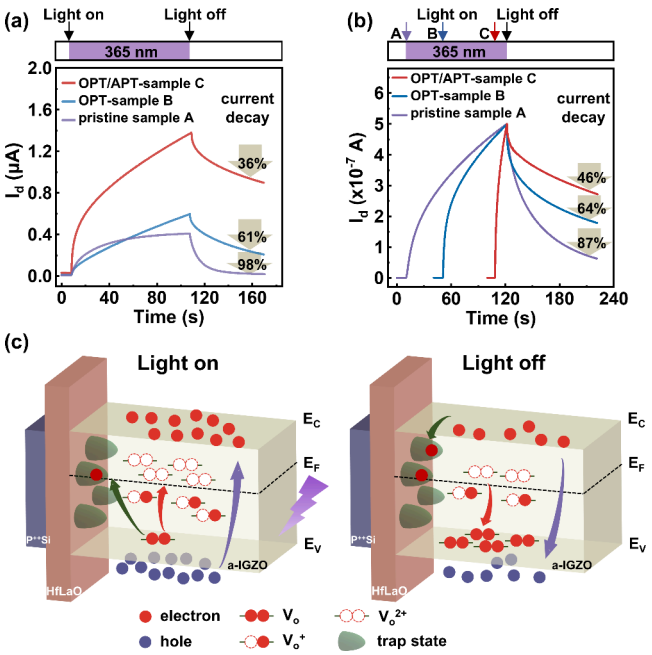


Figure 3. Typical photocurrent under UV illumination and decaying characteristics of pristine sample A, OPT-sample B, and OPT/APT-sample C under (a) the same illumination time (100 s) and (b) the same activation current level (0.5 μA). Reading voltage is set as V_d = 1 V with 0 V gate bias. (c) Proposed schematic of the underlying mechanism for the optical response in a-IGZO TFTs.

with the same illumination time and same activation current level, respectively, during which the drain current is measured at a 1 V V_d with no gate voltage. Under the same illumination time of 100 s, the photocurrent is raised from 0.4 μA for the pristine sample A to 1.4 μA for the OPT/APT-sample C, which shows the steepest slope. As a result, the decay percentage of the photocurrent within 60 s is reduced from 98% to 36% once the light exposure is removed. Moreover, the percentage of current decay within 100 s at the same current level of 0.5 μA is also reduced from 87% to 46% after removing the light, which indicates an enhanced PPC effect of the OPT/APT-sample C. Notably, the illumination time to activate the devices to the same 0.5 μA level is obviously reduced from 110 s for pristine sample A to 13 s for the OPT/APT-sample C, which suggests a reduced optical power consumption. As proposed in Figure 3c, this improved PPC is attributed to the increased donor-like oxygen vacancies in a-IGZO by APT, which can be effectively activated to raise the photocurrent. Meanwhile, a longer time is required for the recombination of the excessive photogenerated carriers after removing the UV light because the interfacial traps to capture electrons are effectively suppressed by the OPT, as shown in Figure S3 (Supporting Information).

Furthermore, the effects of OPT and APT on device performances have been also investigated by fabricating the devices with only OPT-HfLaO (namely, the OPT-sample) and APT-IGZO (namely, APT-sample). The electrical and optical performances of the OPT-sample and APT-sample are shown in Figure S8 (Supporting Information), where APT can improve the photocurrent level under the same illumination time and reduce the illumination time under the same current level. On the other hand, OPT can reduce the decay percentage of the photocurrent after removing the illumination so that sample C with both OPT and APT presents excellent electrical and optical characteristics. Importantly, the OPT/APT-sample C also demonstrates high stability, as reflected by the measurements of common I–V, drain bias stress, bias illumination stress, and gate bias illumination stress after its 1-year exposure to air (Figures S16–S19, Supporting Information) so that the OPT/APT-sample C is selected for further synaptic investigation.

Figure 4a illustrates the operation principle of biological synapses in the human visual system, and the photoreceptors can perceive external lights to produce stimulus signals for synapses, synaptic weights of which can be modulated accordingly. Analogously, the source and drain electrodes in the a-IGZO TFTs are treated as pre- and postsynaptic terminal, respectively. Representing synaptic weight, the conductance of the a-IGZO channel layer is modulated by UV light pulses (365 nm) as external stimuli, which trigger optical synaptic behaviors through channel carriers regarded as biological neurotransmitters. In addition, voltage signals applied at gate electrode are considered as biological neuromodulators, which can effectively affect the activation of neurotransmitters.[13,14]

As a typical short-term synaptic plasticity, paired-pulse facilitation (PPF) stimulated by a UV signal is successfully emulated in the a-IGZO TFT with OPT and APT (sample C), which is of great significance in decoding temporal information in biological systems.[49,50] As shown in Figure 4b, the paired pulses are composed of two successive UV pulses with a 1 s width. With a 1 s paired-pulse time interval (Δt), the photocurrent stimulated by the latter pulse (A_2) is obviously larger than the former one (A_1), producing a 1.6 PPF index defined as A_2/A_1. However, the PPF index monotonously decreases down to 1 with increasing time intervals because the electron–hole recombination is enhanced with longer intervals, hardly maintaining excessive carriers generated by

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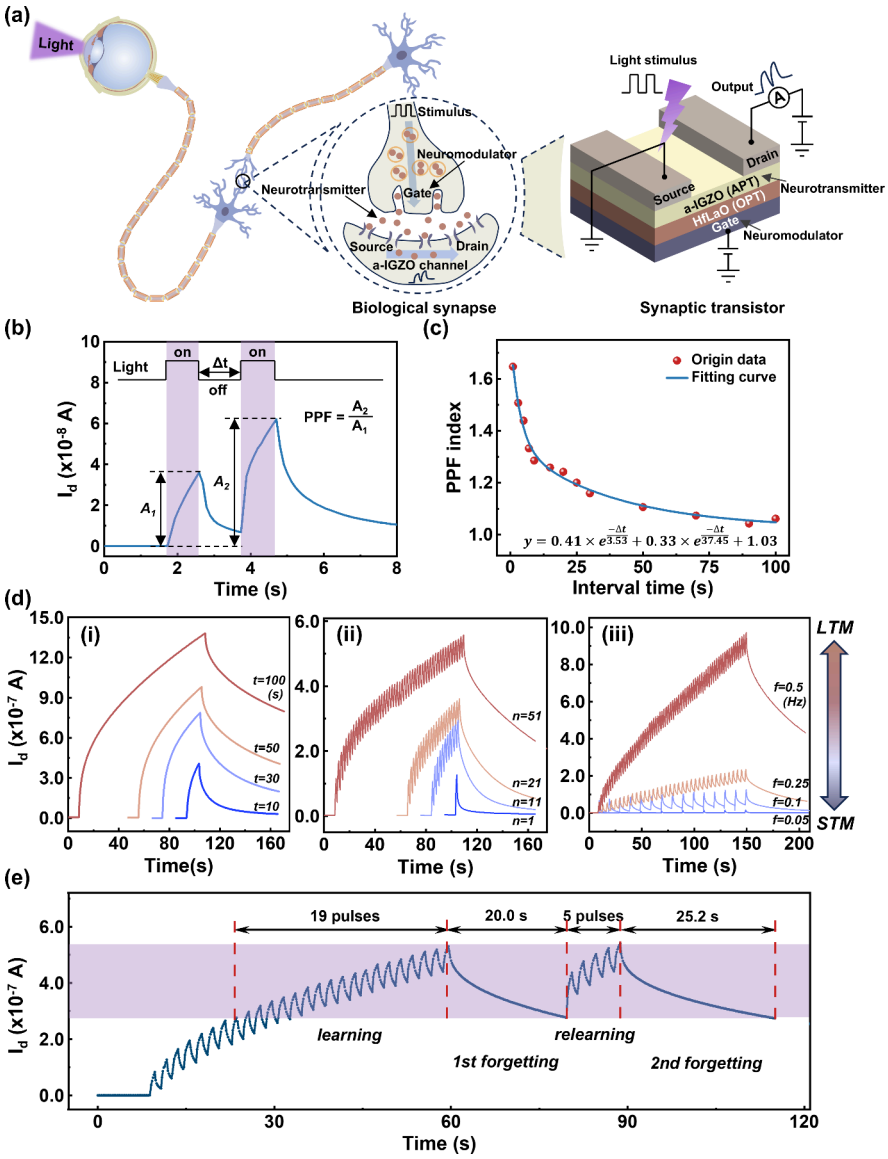


Figure 4. (a) Schematic emulation of the a-IGZO TFTs on an optical synapse in a biological visual system. Simulation of synaptic behavior in the OPT/APT-sample C: (b) A paired-pulse facilitation (PPF) behavior induced by a paired 365 nm UV light pulse with a time interval of 1 s; (c) PPF index as a function of pulse pair interval; (d) The transitions from STM to LTM induced by varying (i) the duration of UV illumination, (ii) the number of UV pulses, and (iii) the frequency of UV pulses; (e) Learning experience behavior by using a sequence of UV pulses (365 nm, 1 s pulse width, 1 s pulse interval). Reading voltage is set as V_d = 1 V with 0 V gate bias.

the former optical pulse. The dependence of the PPF index on the Δt can be well fitted by the following double-exponential equation:[11,51]

$$\mathrm{PPF} = c_1 \cdot e^{\left(-\frac{\Delta t}{\tau_1}\right)} + c_2 \cdot e^{\left(-\frac{\Delta t}{\tau_2}\right)} + c \quad (1)$$

where the two exponential terms describe the rapid and slow decay process; τ_1 and τ_2 represent the characteristic relaxation time of the rapid and slow decay process; c_1 and c_2 are for the initial facilitation magnitude; c is the offset value of the PPF index and Δt is the paired-pulse interval time. As fitted in Figure 4c, parameters are obtained as c_1 = 0.41, c_2 = 0.33, c = 1.03, τ_1 = 3.53 s, and τ_2 = 37.45 s. The value of τ_2 is about one magnitude larger than τ_1, which is consistent with the PPF decaying process in a biological synapse.[10]

Biologically, the "learning−forgetting−relearning" behaviors in human brain are performed by the transitions from short-term memory (STM) to long-term memory (LTM), which is typically achieved by continuous and strong stimuli.[52] The conversion from STM to LTM of the a-IGZO optical synaptic device can be achieved by modulating the durations, amounts, and frequencies of the UV stimuli pulses. As shown in Figure 4d(i), the photocurrent clearly rises with increasing duration of the optical stimuli pulses. With removal of UV light, the drain current decays to the initial level for a short duration to indicate an STM process. On the contrary, the drain current remains at a much higher level than the initial state under prolonged pulse duration, meaning a conversion from STM to LTM. Moreover, it is always required to set the device to the same initial state before applying the light stimuli, which can be achieved by applying a positive voltage pulse at gate electrode for erasure.[53] Furthermore, optical spiking trains with various pulse numbers and frequencies are also investigated for synaptic behaviors. As shown in Figure 4d(ii) and d(iii), the photocurrent is gradually promoted with increasing pulse

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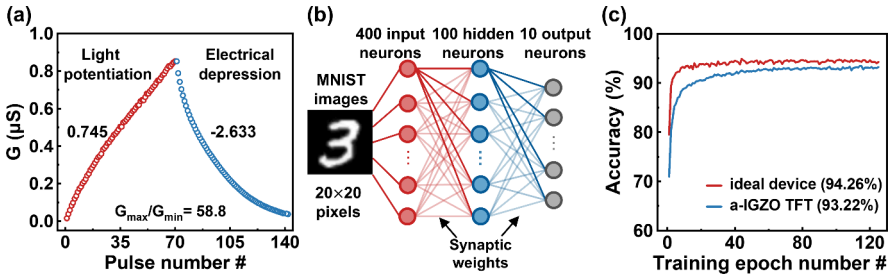


Figure 5. (a) Conductance update by light and electric pulses for potentiation and depression, respectively. Reading voltage is set as V_d = 1 V with 0 V gate bias. (b) Schematic of a two-layer neural network for recognizing handwritten digits. (c) Recognition accuracy as a function of the training epoch.

number and frequency, which are simultaneously accompanied by a slowed-down decay process to convert the STM to LTM. The transitions from STM to LTM of pristine sample A and OPT-sample B were also measured, as shown in Figure S11 (Supporting Information).

In addition, two sequences of UV pulses with a fixed time interval are applied to the device, demonstrating the "learning−forgetting−relearning" function. As depicted in Figure 4e, 19 UV pulses with 1 s width and 1 s interval are first applied to the device and the drain current gradually increases from the initial state to 0.53 μA, representing a learning process. Subsequently, the drain current decays to 0.27 μA within 20 s after the UV stimulation is removed, mimicking a forgetting process. After that, the drain current can be recovered to 0.53 μA by only five pulses, which is much fewer than those required for initial learning to reflect a relearning process. Notably, the decay time after relearning is enlarged from 20 to 25.2 s, which indicates that information is difficult to be forgotten by continuous learning behavior.

During signaling between synapses in the biological nervous system, synaptic weights are dominantly modulated by the neurotransmitters dependent on external stimuli, namely, synaptic plasticity. Interestingly, the activation of neurotransmitters can also be significantly affected by the surrounding environment, such as neuromodulators like astrocytes.[7,8] In this work, biological neuromodulation behavior is concurrently mimicked by adding gate voltage as neuromodulators, as shown in Figure S12 (Supporting Information). With a 1 V gate voltage, the drain current greatly varies under multiple UV pulses, while the modulation of photoconductivity is obviously weakened at a −1 V gate voltage. Therefore, the activation of neurotransmitters that regulated by neuromodulators is successfully emulated by exploiting the inherent three-terminal structure of the a-IGZO TFT.

Importantly, linear and symmetric synaptic plasticities of long-term potentiation and depression (LTP/LTD) are essential to construct neural networks with high accuracy and reliability.[1,54,55] Representing synaptic weight, the channel conductance of the a-IGZO TFT is improved by continuous UV pulses for the synaptic potentiation process. On the other hand, electrical gate voltage pulses are adopted to lower the channel conductance for synaptic depression. The nonlinearity factor (ν), which illustrates the nonlinear behavior of the weight update, is calculated based on the normalized conductance in the potentiation (G_p) and depression (G_d) processes as a function of pulse number (p):[56]

$$G_p = G_{min} + B(1 - e^{-\nu p}) \quad (2)$$

$$G_d = G_{max} - B(1 - e^{\nu(p - p_{max})}) \quad (3)$$

$$B = \frac{G_{max} - G_{min}}{1 - e^{-\nu \cdot p_{max}}} \quad (4)$$

where G_{max} and G_{min} represent the maximum and minimum conductance in the potentiation and depression processes; p_{max} is the maximum pulse number.

As a nearly linear and symmetric weight update can be realized by encoding the input pulses,[57] the dependences of the device conductance on the programming pulse width and amplitude for depression process were systemically studied, as shown in Figure S13 (Supporting Information). As a result, the extracted nonlinearity factor for the electrical depression process is reduced to −2.623 under continuous gate voltage pulses of 1 V with a 0.08 s width and 6.5 s interval, and the optical potentiation nonlinearity is 0.745 under continuous UV light pulses with a 1 s width and 1 s interval. An asymmetric index of 3.368 can be obtained by the difference in the nonlinearity index between LTP and LTD processes. Meanwhile, the a-IGZO TFT exhibits multilevel conductance states up to 70 and a reasonable 58.8 maximum/minimum conductance (G_{max}/G_{min}) ratio, as shown in Figure 5a. Cycled LTP/LTD operations and retention characteristics of the multiconductance state during the LTP process after a 1-year exposure to air were also measured, as shown in Figure S14 (Supporting Information).

Accordingly, an artificial neural network (ANN) for a multilayer perception (MLP) is simulated using the open-source code NeuroSimV3.0.[58] The MNIST data set of black-and-white handwritten digit patterns with a size of 20 × 20 (edge cropped) is used as the training and testing data set. The network topology is 400 (input layer)-100 (hidden layer)-10 (output layer), and each neuron is fully connected via artificial synapses according to the measured nonlinear parameters, as displayed in Figure 5b. Four hundred neurons of input layer correspond to a 20 × 20 MNIST image, and 10 neurons of the output layer correspond to 10 classes of digits. As a result, the recognition accuracy based on the LTP/LTD curve has reached 91.76% after 30 training epochs. The maximum accuracy reaches a high value of 93.22% within the 125 training epochs, which is close to the 94.26% that was obtained from an ideal synapse, as shown in Figure 5c. The accuracy difference between the ideal 94.26% and practical 93.22% should be attributed to the nonideal linearity, symmetry, and dynamic range of conductance in the as-fabricated a-IGZO optical synaptic transistor for which fabrication methods like ambience modulation and postdeposition annealing should be further optimized. Additionally, a comparison of our a-IGZO optical

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Table 2. Comparison of Properties with Previously Reported Optical Synaptic Devices[a]

					potentiation/depression				recognition task		
active layer	method (T_{max})	mobility ($cm^2/V{\cdot}s$)	array	PPF index	stimuli (P/D)	state #	G_{max}/G_{min}	NL (P/D)	epoch	accuracy (%)	ref
IGZO-alkylated GO	SC (80 °C)			1.82	light (405 nm) + electric/electric	100	32	3/7	60000	62	51
$CsPbBr_3$ QDs/PDVT-10	SC (90 °C)	0.15		2.74	light (405 nm)/electric	50	<20		37	89.98	59
$Cs_2AgBiBr_6$/IGZO	space-confined/PLD (200 °C)	<12		2.06	light (455 nm)/electric	600	<45	>1/<3	200/18000	83.8/89.3	60
pentacene/PVP	TE/SC (200 °C)			1.13	light (365 nm)/electric	30	<3		100	88	61
CDs/silk/pentacene	SC/TE (60 °C)			1.24	light (365 nm)/electric	100	15.6		15000	73	62
UCNPs/MoS_2/pentacene	SC/TE (100 °C)			1.9	light (980 nm)/electric	100	15.5	3.4/5.9	15000	65	63
$MoSe_2/Bi_2Se_3$/PMMA/pentacene	SC/TE (120 °C)			1.53	light (790 nm)/electric	100			1500	90	64
CuPc/p-6P	thermal deposition (180 °C)				light (365 nm)/electric	85	48.2	1.38/−6.11	60000	78	65
IGTO	sputtering (700 °C)				light (467 nm)/electric	64	6.58	1.13/2.03	120	90.42	66
TiO_2/IGZO	sputtering (400 °C)	1.32		1.85	light (520 nm)/electric	50		0.395/−4.6	92	90.3	67
a-IGZO	sputtering (RT)	20.8	5 × 5	1.6	light (365 nm)/electric	70	58.8	0.75/−2.63	30/125	91.76/93.47	this work

[a]GO: graphene oxide; PDVT-10: poly[2,5-bis(alkyl)pyrrolo-[3,4-*c*]pyrrole-1,4(2*H*,5*H*)-dione-*alt*-5,5′-di(thiophen-2-yl)-2,2′-(*E*)-2-(2(thiophen-2-yl)vinyl)thiophene]; QDs: quantum dots; PVP: poly(4-vinylphenol); CDs: carbon dots; UCNPs: upconverting nanoparticles; PMMA: poly(methyl methacrylate); CuPc: copper-phthalocyanine; p-6P: *para*-sexiphenyl; T_{max}: maximum temperature during fabrication; SC: spin coating; PLD: pulsed laser deposition; TE: thermal evaporation; RT: room temperature; PPF: paired-pulse facilitation; G_{max}/G_{min}: ratio of maximum conductance to minimum conductance; NL: nonlinearity.

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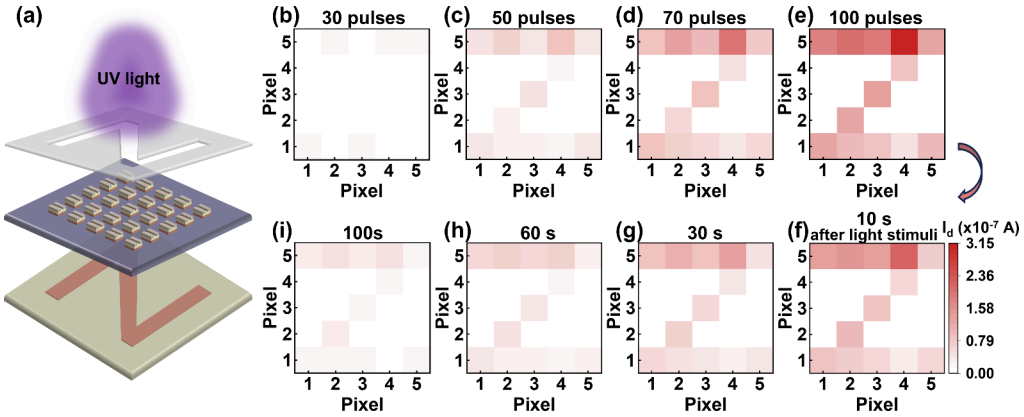


Figure 6. (a) Schematic diagram of a 5 × 5 a-IGZO optical synaptic transistor array for visual recognition and memory. (b–e) The image mapping of an array illuminated by 30, 50, 70, and 100 UV pulses at V_g = 0 V. (f–i) The image mapping of an array after removing the light for 10, 30, 60, and 100 s. Reading voltage is set as V_d = 1 V with 0 V gate bias.

synaptic transistor to previously reported devices is included in Table 2.

Last but not least, 5 × 5 perceptive hardware based on the a-IGZO optical synaptic transistors has been successfully fabricated as illustrated in Figure 6a. The array is illuminated by a 365 nm UV light through a mask with a "Z" letter shape for image recognition. As shown in Figure 6b–i, the drain current of each pixel is recorded and presented in the form of charts, which are obtained for the UV stimuli pulses with numbers of 30, 50, 70, and 100. After the illumination of 100 UV pulses, the "Z" pattern can be clearly perceived, which gradually becomes blurred with an increase of decay time. In addition, the effect of gate voltages on the image recognition is also studied as displayed in Figure S15 (Supporting Information), where a clear "Z" image is obtained at a 1 V gate voltage but a blurred and indistinguishable image at a −1 V gate voltage. After the UV was removed over 100 s, the recognized image at 1 V gate voltage remains relatively distinct while that at −1 V gate voltage disappears. This suggests that the recognition and memory levels of the optical inputs can be modulated with different gate voltages, which is similar to the modulation processes of neuromodulators in the nervous system. Meanwhile, this electrically assisted operation suggests a ready modulation of possible memory retention associated with enhanced PPC.

3. CONCLUSION

In summary, a-IGZO optical synaptic transistors with high-κ HfLaO as the gate dielectric have been successfully fabricated at room temperature. Thanks to the postdeposition plasma treatments of Ar on a-IGZO and O_2 on HfLaO, the a-IGZO TFTs demonstrate a high carrier mobility of 20.8 $cm^2/V{\cdot}s$, a high I_{on}/I_{off} ratio of 3.2 × 10^6, and a small subthreshold swing (SS) of 0.25 V/dec. Moreover, the PPC effect is successfully enhanced, as reflected by the reduction of current decay percentage from 98% to 36% under the same illumination time and from 87% to 46% under the same activation current level. These high performances can be attributed to the increased donor-like oxygen vacancies in the APT a-IGZO and suppressed oxygen vacancy defects in the OPT HfLaO. Accordingly, optical synaptic functionalities are achieved with concurrent emulations of neurotransmitters and neuromodulators. Furthermore, highly linear and symmetric conductance updating characteristics are achieved by encoding the input pulse, which supports a 93.22% accuracy to recognize handwritten digits in the MNIST data set. Importantly, optical synaptic hardware has been fabricated based on the a-IGZO TFTs to simulate the biological vision system, which can accurately recognize a letter pattern to demonstrate potentials for future neuromorphic applications. Further refinements on the integration process can be explored by adopting passivation technology and thermal treatments to further improve the stability and reliability of array-level integration in practical applications.

4. EXPERIMENTAL SECTION

Fabrication of a-IGZO TFTs. First, p^{++} Si wafers were successively dipped in ethanol and deionized water, each for 15 min with ultrasound. Then, the HfLaO gate dielectric films were deposited by cosputtering a Hf metal target at 12 W DC power and a La metal target at 45 W DC power in an $Ar/N_2/O_2$ (24 sccm/1 sccm/6 sccm) ambience at 1 Pa chamber pressure. Then, the samples were divided into three groups, named samples A–C. Samples B and C were treated by O_2 plasma at 50 W for 1 min. Subsequently, the a-IGZO films were deposited on all samples simultaneously by cosputtering an In metal target at 4 W DC power and an IGZO ceramic target ($In_2O_3/Ga_2O_3/ZnO$ = 1:1:1) at 60 W RF power in an Ar/O_2 (30/0.7 sccm) ambience at 1 Pa chamber pressure. Next, only sample C was treated by Ar plasma at a power of 50 W for 1 min. Moreover, the target/substrate distance, sputtering gun angle, and sample tray rotation speed were set as 10 cm, 70°, and 10 rpm (revolutions per minute), respectively, during the sputtering for promoting film uniformity and quality. Finally, Cr (10 nm) and Ag (50 nm) were deposited on all samples by thermal evaporation through a shadow mask to define the 40 μm channel length and 300 μm channel width. Moreover, Cr/Ag/HfLaO/Si capacitor structures were also fabricated at the same run with corresponding TFTs.

Device Characterization and Measurement. The chemical binding states of HfLaO and a-IGZO thin films were investigated by XPS (Thermo escalab 250XI), and the surface morphology was characterized by using the AC Air Topography mode in AFM (Asylum Research Cypher S). The electrical and optical characteristic measurements of the a-IGZO TFTs were conducted by using a Keithley 2634B semiconductor analyzer at room temperature. The light

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sources used in the measurement are LEDs with wavelengths of 365, 460, 520, and 620 nm. The dielectric capacitance is measured by the C−V model of a Keysight B1500 semiconductor parameter analyzer.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsphotonics.4c01809.

XPS spectra of In 3d and AFM images for pristine and Ar-plasma-treated a-IGZO films. XPS spectra of O 1s and AFM images for pristine and O_2-plasma treated HfLaO films. The contact resistance extracted by TLM. Transfer hysteresis characteristics of samples. C−V characteristics. Electrical and optical characteristics of pristine sample, APT-sample and OPT-sample. The variations of electrical characteristics with morphology improvement and indium chemical changes. Transitions from STM to LTM of pristine sample A and OPT-sample B. Facilitation behaviors modulated by gate voltage. Linearity optimation of weight updating in depression process. Cycled LTP/LTD operations and retention characteristics of multiconductance state. Image mapping of a-IGZO TFTs array under different modulating voltages. Stability measurements for OPT/APT-sample C after its 1-year exposure to air (PDF)

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Notes

The authors declare no competing financial interest.

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